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To cite this article: Ronald Buliung (2025) 15-minute cities, 'walkability' and last millimeter problems, *Disability & Society*, 40:5, 1425-1429, DOI: [10.1080/09687599.2024.2385919](https://doi.org/10.1080/09687599.2024.2385919)

To link to this article: <https://doi.org/10.1080/09687599.2024.2385919>



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Published online: 02 Aug 2024.



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15-minute cities, ‘walkability’ and last millimeter problems

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ABSTRACT

The 15-minute city concept is about planning and building cities so that most essential activities can be accessed within 15-minutes of a residence on foot or by bicycle. The possibilities and limitations of the concept for disabled persons have yet to be fully considered. Here I centre disability in a hopeful critique of concepts central to 15-minute city implementation. I argue that greater attention should be given to what I call last millimeter problems, barriers within small spaces (e.g. subway platform gaps) disabled persons face in transportation systems and the built environment that can make any city, including a 15-minute city, an impassable city.

ARTICLE HISTORY

Received 31 March 2024

Accepted 15 July 2024

KEYWORDS

15-minute city; x-minute city; disability; ableism; walkability; walking; barriers

Introduction

As far as urban planning concepts go, the 15-minute city has become somewhat of a global phenomenon. Urbanist Carlos Moreno is typically credited with conceiving of this latest iteration of the x-minute city. The x-minute city idea, where x refers to minutes of travel to reach essential centers or amenities, has been around for a while. Moreno’s public works begin around 2016 with an academic article arriving later. The concept is described as, ‘an urban set-up where locals are able to access all of their basic essentials at distances that would not take them more than 15min by foot or by bicycle’ and is thought to hold promise for producing healthy, sustainable and inclusive urban futures (Moreno et al. 2021). Having what you need close by is not a terrible idea, and certainly could benefit many disabled persons who often face temporal problems with urban transportation (e.g. Bezyak, Sabella, and Gattis 2017; Galer, 2023). That said, disability is entirely absent from the foundational 15-minute conceptual texts and negligibly considered in related positivist or post-positivist proof of concept studies (e.g. Capasso Da Silva, King,

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and Lemar 2019; Weng et al. 2019; Willberg, Fink, and Toivonen 2023; Yu and Higgins 2024).

Reaction to the 15-minute city concept includes enthusiasm from many planners and other urbanists and an unproductive conspiracy minded focus on possible threats to individual liberty and automobiles (Patterson and Barrie 2023). In this commentary I center disability in a critique of 15-minute city rhetoric and research. Using a critical disability studies lens to work through the concept is a necessary intervention. Like Khavarian-Garmsir, Sharifi, and Sadeghi (2023), I think about how the concept hints at physical determinism and a lack of attention given to the how (will we build it), what (will it look like) and for whom (who is included) aspects of the conversation. I am less fixated on the 15-minutes per se and more concerned about what things might look like and how they might work (or not) when they are built. I consider the invisibility of disability and the application of ableist conceptualizations of mobility like 'walkability' and 'walk scores', concepts and methods that appear central to x-minute thinking. I draw on a history of disability and urban transportation problems to illustrate how cities can become impassible in the presence of small scale, big impact, limitations and failings in infrastructure and services.

The problem with 'walkability'

Walkability has been defined as the manner in which land use and the built environment coalesce to support or encourage walking for recreation, work, or to access services (Leslie, Butterworth, and Edwards 2006). Walkability is often assessed using walk scores; comprehensive measures of neighbourhood walkability based on shortest distances to destinations, block length, intersection density, and sometimes population density (Hall and Ram 2018; WalkScore.com). Walk scores affect housing prices (Gilderbloom, Riggs, and Meares 2015). Walkability and walk scores are ableist constructs.

According to the Oxford English Dictionary 'to walk' means to:

'move at a regular pace by lifting and setting down each foot in turn, never having both feet off the ground at once'.

According to walkscore.com, my home's walk score is 98/100. I basically live in a 15-minute walkable neighbourhood; I like it but it is not a universally accessible neighbourhood. WalkScore.com tells me my 'location is a Walker's Paradise so daily errands do not require a car'. This 'paradise' assumes one 'walks' according to plan. By giving primacy to a bipedal universal conceptualisation of locomotion and walking bodies, walk scores are ableist. They also provide a partial and ableist representation of neighbourhood accessibility.

My daughter uses a power wheelchair. She uses the term ‘walking’ to describe her neighbourhood travel. Despite living in ‘paradise’ my daughter cannot get into (i.e. stepped entry, narrow door width, interior obstacles) all of the places where she wants or needs to go including the nearest school. Walk scores stop at the door offering no information about doorways conditions (i.e. thresholds), internal navigation of destinations, or infrastructure surface conditions (e.g. cracked or sloping sidewalks, etc.).

If one of the key quantifiable pillars of the walk score is distance to destinations, what happens to a walk score when we start to eliminate destinations that are entirely inaccessible to wheelchair users? What happens to the walk score if we embed within it, the diversity of disability? Numerically speaking, my paradise might begin to look more like an accessibility desert.

Indeed, what’s the point in having something close by if you can’t even get in the door?.

As for schools, Moreno’s original paper mentions schools and school playgrounds several times. An inaccessible neighbourhood school is 550 metres away. My daughter’s assigned school is 4.1 km away. Google Maps tells me the school trip is 8 min by car, 29 min by transit, 44 min by walking and 12 min on a bike. My daughter’s 15-minute city does not include a school because of historical (i.e. the closest school was built in the nineteenth century) and contemporary intersections between ableism and education policy.

Walkability, walking in the 15-minute city, and walk scores, are very much focused on the moving around on two feet. Attention is primarily given to door-to-door analysis, rather than ‘can I get out this door and then in that door at the other end of my trip?’ analysis. The latter problem requires a sort of embodied analysis that is missing from x-minute discourse and research.

Last mile, last millimetre, 15-minute(s) late ... or perhaps not at all

The urban transportation problems faced by disabled persons are well documented. Consider paratransit, i.e. door-to-door service for disabled transit users. As Galer notes, ‘Ever since paratransit services were introduced by the Toronto Transit Commission in 1975, there were problems’ (Galer, 175, 2023). Problems include employment discrimination; vehicles arriving late or not at all; schedule complexity; cost; and even injury and death (Galer, 2023). In 1980, Lynne Pyke died while exiting a Wheeltrans vehicle in Toronto, in 2023 Guillermo Aviles died in Los Angeles following a drop off. Yes, people die using public transit, but the death of disabled persons arguably arises through ableist causal mechanisms. The urban transportation problems of disabled persons are troubling, unnecessary, costly, and historically persistent (e.g. Bezyak, Sabella, and Gattis 2017; [removed for anonymity]; Sitter and Mitchell 2020).

When I think about these issues, I think about what I call last millimeter problems, i.e. limitations or failures of design and legislative compliance and the poor

physical condition of infrastructure (mobile or fixed) within small areas that can turn any city, including a 15-minute city, into an impassable city. Some of these problems include: door widths; subway platform gaps; stepped-up curbs and doorways; missing surfaces that provide guidance or warning to Blind pedestrians through tactile feedback; door actuator button obstruction or height; excessive ramp or sidewalk slope; loose carpets; non-permitted use of disabled parking spots; vegetation encroachment onto sidewalks; ice and snow accumulation on sidewalks; and so on. Trips can end before they even begin.

My daughter's chair has become lodged in the horizontal gap between a subway's platform and car. The Toronto Transit Commission accepted the recommendation of its Advisory Committee on Accessible Transit regarding acceptable 89mm and 38mm for horizontal and vertical gaps respectively; last millimeter problems indeed. A partial retrofit campaign will cost at least \$28 million.

What about time? When does the 15-minute clock start? Does it include the extra time disabled persons spend on trip planning irrespective of the travel mode? Does it include care time; the time before leaving that can involve complex preparation and care (Stevens 2018)? Or the time it might take some people with a physical disability to move from place to place? And what happens when my 15-minutes falls apart because a transit vehicle is excessively delayed or a school bus doesn't show up at all? Plans made then unmade by the systemic failure to provide reliable service to disabled persons.

Fifteen-minute city thinking seems largely focused on the important problem of getting 'to' things, with less attention given to experiences along the way or the problem of getting across a threshold at the beginning or end of a trip. I am reminded of a piece by Koch (2008) where he described a conceptual rift between geographical accessibility research and disability studies. The former tends to dominate 15-minute studies and includes a sort of disembodied analysis of the layout of cities and transport connectivity. The latter tends to focus on the lived experiences of disabled persons offering a necessary understanding of the barriers disabled persons face in cities. Indeed, there are experiential, legal, economic and intellectual consequences associated with a failure to explicitly include disability in city building discourse and scholarship from the beginning. If 15-minute urbanism is going to even come close to its supposed promise, particularly regarding the production of healthy and inclusive urban futures, then more attention must be given to the how, what and for whom questions. The privileging of non-disabled bodies in x-minute rhetoric and proof of concept research cannot continue. The invisibility of disability in this work could have terrible knock-on effects as rhetoric becomes plans, and plans become the material urban settings of everyday life.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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